Lesson 2.5 Teaching Support Guide

Teach-tomorrow route

Print: current Guided Notes and Practice A for the class; Exit Tickets; optional Language Support and supported homework. Keep this guide and answer keys teacher-facing. Use the full-size model pages in the updated notes.

Phase	Planned minutes	Teaching move
Retrieve	5	Warm-Up items 1–4; note the missing prerequisite before launch.
Launch	7	Name the session target and invite a prediction from the conceptual task.
Model	10	Link a visual model to the worked example; rehearse the mathematical explanation.
Practice	18	Core: A #1, #2, #4, #6 and the error task. Choose remaining items from evidence.
Check	5	Exit Ticket; use the diagnostic page before assigning reteaching.

This is a planned 45-minute route, not a measured classroom duration. For a shorter lesson, keep the model and exit; reduce repeated practice. Form B is parallel practice or reassessment only when its items are unseen.

S1: Range and IQR · I can find the range and the interquartile range of a data set.

S2: Comparing Variability · I can compare the variability of two data sets using range and IQR.

Spaced retrieval: revisit one item after 2 lessons and after approximately 1 week. Mix it with the next unit’s prerequisite check; do not announce the operation before students choose a method.

Session 1 · Range and IQR

I can find the range and the interquartile range of a data set. · 6.SP.B.5c

Observe and respond

Exit item	Expected evidence	Next prompt/model
1	Order: 3, 5, 7, 8, 10, 12, 14. a. 11 b. 12 − 5 = 7	Recompute the two quartiles and identify which observations determine them before making a claim.
2	30 − 6 = 24; 22 − 10 = 12	Recompute the two quartiles and identify which observations determine them before making a claim.
3	B — Only the range changes	Ask the learner to show why B — Only the range changes is correct and reject a specific distractor using a model.

Distinguish the error

Conceptual signal: Treats resistance to extremes as a rule that IQR never changes..

Response: Recompute the two quartiles and identify which observations determine them before making a claim.

Also check: Computing the range (max − min) when the IQR is asked.

Also check: Subtracting Q1 from the max, or the min from Q3, instead of Q3 − Q1.

If calculation alone is wrong, use Session 1 Prerequisite Practice A for the matching skill, then return to the model. If the interpretation is wrong, label the quantities before reteaching the operation. Independent re-entry requires a correct setup and a connected explanation.

Session 1 · Fresh check and scoring

Reserve this check until after revision; it is distinct from the exit ticket. Do not show this teacher page to learners beforehand.

Compare IQRs for 2, 4, 6, 8 and 2, 4, 6, 20. Explain the effect of changing the maximum.

Teacher solution: First: Q1 3, Q3 7, IQR 4. Second: Q1 3, Q3 13, IQR 10. It changes because the maximum is part of the upper-half median.

Score the evidence

Score	Evidence	Next action
3	Valid representation, correct result/units, connected explanation.	Continue; fade a support next time.
2	Correct setup and meaning with a calculation/notation slip, or an incomplete explanation.	Brief targeted feedback, then another unseen check.
1	Some relevant facts but a model or interpretation error.	Use the named model and the corresponding prerequisite page.
0	No interpretable evidence yet.	Oral/diagram response to separate access needs from mathematics.

Complete response anchor: A: range 3, Q1 1.5, Q3 3.5, IQR 2. B: range 39, Q1 1.5, Q3 21.5, IQR 20. No; in these small sets the maximum participates in Q3. The seven-value examples have different quartile positions.

Misconception response anchor: Treats resistance to extremes as a rule that IQR never changes.. Use recompute the two quartiles and identify which observations determine them before making a claim.

Language access: point/draw and label → oral rehearsal → relationship sentence → independent explanation. Offer the support needed for this task and fade it based on evidence. Suggested frame: The IQR changed from ___ to ___ because Q3 ___. This shows that ___.

Session 2 · Comparing Variability

I can compare the variability of two data sets using range and IQR. · 6.SP.B.5c

Observe and respond

Exit item	Expected evidence	Next prompt/model
1	a. 12 b. 4 (Y is more consistent)	Make two comparison rows: range versus range, and IQR versus IQR. Attach the purpose to each row.
2	Team 1 (IQR 6 < 14)	Make two comparison rows: range versus range, and IQR versus IQR. Attach the purpose to each row.
3	B — IQR	Ask the learner to show why B — IQR is correct and reject a specific distractor using a model.

Distinguish the error

Conceptual signal: Compares one group's range with another group's IQR or calls a group simply 'more spread out' without a measure..

Response: Make two comparison rows: range versus range, and IQR versus IQR. Attach the purpose to each row.

Also check: Comparing the range of one group with the IQR of the other.

Also check: Thinking a greater IQR always means a greater range.

If calculation alone is wrong, use Session 2 Prerequisite Practice A for the matching skill, then return to the model. If the interpretation is wrong, label the quantities before reteaching the operation. Independent re-entry requires a correct setup and a connected explanation.

Session 2 · Fresh check and scoring

Reserve this check until after revision; it is distinct from the exit ticket. Do not show this teacher page to learners beforehand.

X has range 30 and IQR 4; Y has range 18 and IQR 8. Which is less spread out overall, and which has the steadier middle half?

Teacher solution: Y overall by range; X in the middle half by IQR.

Score the evidence

Score	Evidence	Next action
3	Valid representation, correct result/units, connected explanation.	Continue; fade a support next time.
2	Correct setup and meaning with a calculation/notation slip, or an incomplete explanation.	Brief targeted feedback, then another unseen check.
1	Some relevant facts but a model or interpretation error.	Use the named model and the corresponding prerequisite page.
0	No interpretable evidence yet.	Oral/diagram response to separate access needs from mathematics.

Complete response anchor: A: range 38 and IQR 4. B: range 20 and IQR 10. B has the smaller overall spread; A has the smaller middle-half spread. A claim must name the measure and purpose.

Misconception response anchor: Compares one group's range with another group's IQR or calls a group simply 'more spread out' without a measure.. Use make two comparison rows: range versus range, and IQR versus IQR. Attach the purpose to each row.

Language access: point/draw and label → oral rehearsal → relationship sentence → independent explanation. Offer the support needed for this task and fade it based on evidence. Suggested frame: Using ___, I choose ___ because ___. However, the other measure shows ___.